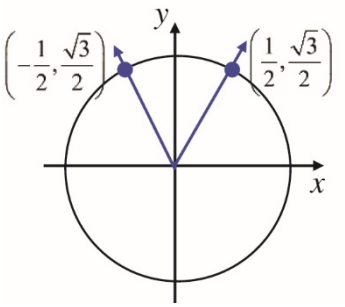
Solve Trigonometric Equations in Linear Form

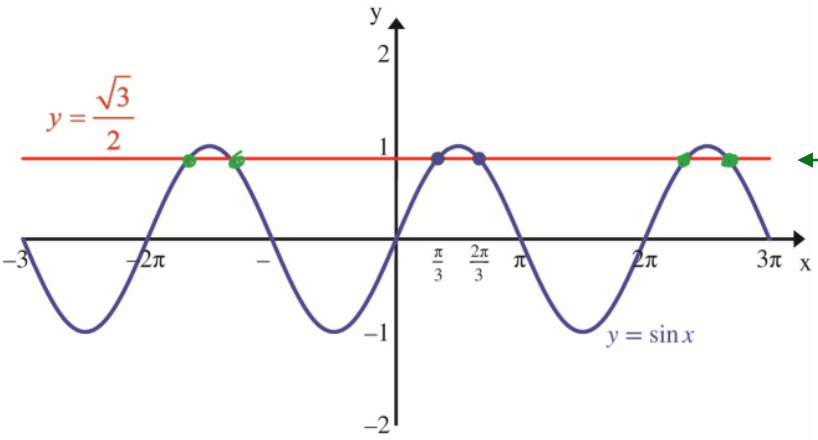
Example: Solve. $\sin x = \frac{\sqrt{3}}{2}$

From the unit circle, we can see that there are 2 values of x between 0 and 2π that will solve the equation: $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$.



However, there are actually infinitely many solutions to the equation:

$x = \frac{\pi}{3} + 2n\pi$ and $x = \frac{2\pi}{3} + 2n\pi$ (where n is any integer).



all intersections are solutions

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1. Solve the equation on the interval $[0, 2\pi)$. $3\csc x = 4 + \csc x$

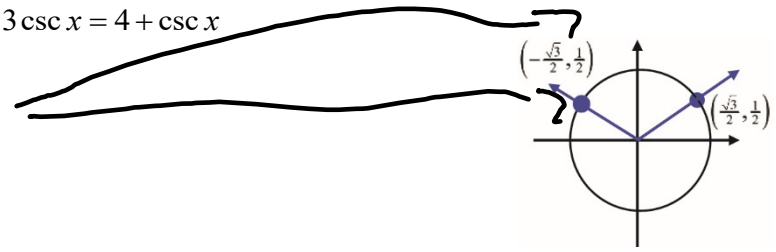
$2\csc x = 4$

$\csc x = \frac{4}{2}$

$\csc x = 2$

$\sin x = \frac{1}{2}$

$x =$



2. Solve the equation over the set of real numbers. $\sqrt{2}\cos x + 5 = 4$

$\sqrt{2}\cos x = -1$

$\cos x = \frac{-1}{\sqrt{2}}$

$x =$

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Solve Trigonometric Equations Involving Multiple or Compound Angles

Solve the equation on the interval $[0, 2\pi)$.

3. $8\cot 2x - 5 = 3$

Isolate $\cot 2x$.

Give the general solution from the unit circle.

To find solutions on the interval $[0, 2\pi)$, substitute $n = \ldots, -1, 0, 1, 2, \ldots$

$8\cot 2x = 8$

$\cot 2x = 1$

$\tan 2x = 1$

$2x = \frac{\pi}{4}, \frac{5\pi}{4}$

$x = \frac{\pi}{8} + n\pi$ & $x = \frac{5\pi}{8} + n\pi$

4. $\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

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Solve Higher-Degree Trigonometric Equations

Solve the equation on the interval $[0, 2\pi)$.

5. $2\sin^2 x + 5\sin x + 3 = 0$

Use Identities to Solve Trigonometric Equations

Solve on the interval $[0, 2\pi)$.

6. sin 2x = cos x

7. sin x = cos x

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Use Inverse Functions to Express Solutions to Trigonometric Equations

8. Solve on the interval $[0, 2\pi)$. $4\cos^2 x - 7\cos x + 3 = 0$

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